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pharmaniaga®

SII PNEUMOSIL™

SUSPENSION FOR INJECTION

Pneumococcal Polysaccharide Conjugate Vaccine (Adsorbed) (10-valent)

BRAND NAME/PRODUCT NAME

SII PNEUMOSIL

SUSPENSION FOR INJECTION (SINGLE DOSE - 1 DOSE)
Pneumococcal polysaccharide conjugate vaccine (adsorbed) (10-valent)

SUSPENSION FOR INJECTION (MULTI DOSE - 5 DOSE)
Pneumococcal polysaccharide conjugate vaccine (adsorbed) (10-valent)

DESCRIPTION

SII PNEUMOSIL [Pneumococcal Polysaccharide Conjugate Vaccine (10-valent, adsorbed)] is a sterile suspension of saccharides of the capsular antigens of *Streptococcus pneumoniae* serotypes 1, 5, 6A, 6B, 7F, 9V, 14, 19A, 19F and 23F individually conjugated by using 1-cyano-4-dimethylamino pyridinium tetrafluoroborate chemistry (CDAP) to non-toxic diphtheria CRM197 protein. The polysaccharides are chemically activated and then covalently linked to the protein carrier CRM197 to form the glycoconjugate.
Individual conjugates are compounded and then polysorbate 20 and aluminium phosphate are added to formulate the vaccine. The potency of the vaccine is determined by the quantity of the saccharide antigens and the saccharide-to-protein ratios in the individual glycoconjugates. The vaccine meets the requirements of WHO and BP when tested by the methods outlined in WHO TRS 977 and BP.

The vaccine is whitish turbid liquid. A fine white deposit with clear colourless supernatant may be observed upon storage of the vial. This does not constitute a sign of deterioration.

COMPOSITION

SII PNEUMOSIL (10-valent) 0.5 ml - 1 dose

Each dose of 0.5 ml contains:

Saccharide for serotypes 1,5, 6A, 7F, 9V, 14, 19A, 19F and 23F	2 mcg each
Saccharide for serotype 6B	4 mcg
CRM 197 (carrier protein)	19 to 48 mcg
Aluminium (as Aluminium phosphate)	0.125 mg

SII PNEUMOSIL (10-valent) 2.5 ml - 5 dose

Each dose of 0.5 ml contains:

Saccharide for serotypes 1,5, 6A, 7F, 9V, 14, 19A, 19F and 23F	2 mcg each
Saccharide for serotype 6B	4 mcg
CRM 197 (carrier protein)	19 to 48 mcg
Aluminium (as Aluminium phosphate)	0.125 mg
Thiomersal (as preservative)	0.005%

Other ingredients: L-Histidine, Succinic acid, Sodium chloride, Polysorbate 20 and water for injections.

INDICATIONS

Active immunization for the prevention of invasive disease, pneumonia and acute otitis media caused by *Streptococcus pneumoniae* serotypes 1, 5, 6A, 6B, 7F, 9V, 14, 19A, 19F and 23F in infants and toddlers from 6 weeks up to 2 years of age. The use of vaccine should be determined on the basis of relevant recommendations and take into consideration the disease impact by age and regional epidemiology.

DOSAGE AND ADMINISTRATION

Method of Administration

For intramuscular injection only.
The dose is 0.5 ml given intramuscularly, with care to avoid injection into or near nerves and blood vessels. The preferred sites are anterolateral aspect of the thigh in infants or the deltoid muscle of the upper arm in young children. The vaccine should not be injected in the gluteal area. Do not administer **SII PNEUMOSIL** (10-valent) intravascularly. The vaccine should not be injected intradermally, subcutaneously or intravenously, since the safety and immunogenicity of these routes have not been evaluated.

Once opened, multi-dose vials should be kept between +2°C and +8°C. Multi-dose vials of **SII PNEUMOSIL** (10-valent) from which one or more doses of vaccine have been removed during an immunization session may be used in subsequent immunization sessions for up to a maximum of 28 days, provided that all of the following conditions are met (as described in the WHO policy statement: Handling of multi dose vaccine vials after opening, WHO/IVB/14.07):

- The vaccine is currently prequalified by WHO;
- The vaccine is approved for use for up to 28 days after opening the vial, as determined by WHO;
- The expiry date has not passed;
- The vaccine vial has been, and will continue to be, stored at WHO - or manufacturer recommended temperatures; furthermore, the vaccine vial monitor, if one is attached, is visible on the vaccine label and is not past its discard point, and the vaccine has not been damaged by freezing.

Instruction for Use

For 1 dose (Single Dose) / 5 dose (Multi Dose)

The product is a suspension containing an adjuvant, shake vigorously immediately prior to use to obtain a homogenous, whitish turbid liquid in the vaccine container.

The vaccine should be visually inspected for any foreign particulate matter and/or variation of physical aspect prior to administration. In event of either being observed, discard the vaccine.

The vaccine should be allowed to reach room temperature before use.

When using a Multi Dose vial, each 0.5 ml dose should be withdrawn using a sterile needle and syringe; precautions should be taken to avoid contamination of the contents.

Vaccination Schedule

SII PNEUMOSIL (10-valent) is to be administered as a three-dose primary series at 6, 10, and 14 weeks of age or 2, 3 and 4 months of age or 2, 4 and 6 months of age, with or without, depending on recommended dosing schedule, a booster dose at 9-10 or 12-15 months of age. The minimum interval between doses should be 4 weeks. If a booster dose is given, it should be at least 6 months after the last primary dose.
Alternatively, **SII PNEUMOSIL** (10-valent) is given as a two-dose primary series with booster dose. The first dose may be administered from the age of 6 weeks, with a second dose at age of 14 weeks. The third (booster) dose is recommended between 9-18 months of age.

Table 1: Vaccination Schedule for Infants and Toddlers

Dosage Schedules	Dose 1[a, b]	Dose 2[b]	Dose 3[b]	Dose 4[c, d]
3p+1	6 weeks	10 weeks	14 weeks	9 - 10 months or 12-15 months
3p+0	6 weeks	10 weeks	14 weeks	–
2p+1	6 weeks	–	14 weeks	9 - 18 months

[a] Dose 1 may be given as early as 6 weeks or at 2 months of age
[b] The recommended dosing interval is 4 to 8 weeks
[c] A booster (fourth) dose is recommended at least 6 months after the last primary dose and may be given from the age of 9 months onwards (preferably between 12 and 15 months of age)
[d] A booster dose is recommended at the age of 9-18 months of age.

For children who are beyond the age of routine infant schedule, the following **SII PNEUMOSIL** (10-valent) schedule is proposed:
The catch-up schedule, for children 7 months through 2 years of age who have not received **SII PNEUMOSIL** (10-valent):

Table 2: Vaccination Schedules for Unvaccinated Children 7 Months of Age Through 2 Years of Age

Age at first dose	Total Number of 0.5 ml doses
7-11 months of age	3[a]
12-24 months of age	2[b]

a. The vaccination schedule consists of two primary doses of 0.5 ml with an interval of at least 1 month between doses. A booster (third) dose is recommended in the second year of life with an interval of at least 2 months after the last primary dose.
b. The vaccination schedule consists of two doses of 0.5 ml with an interval of at least 2 months between doses.

CONTRAINDICATIONS

Hypersensitivity to any component of the vaccine, including diphtheria toxoid.

SPECIAL WARNINGS AND PRECAUTIONS:

RISK OF SENSITIZATION IN RELATION TO THIOMERSAL AND OTHER PRESERVATIVES.

As with all injectable vaccines, appropriate medical treatment and supervision must always be readily available in case of a rare anaphylactic event following the administration of the vaccine.

ADRENALINE INJECTION (1:1000) MUST BE IMMEDIATELY AVAILABLE SHOULD AN ACUTE ANAPHYLACTIC REACTION OCCUR DUE TO ANY COMPONENT OF THE VACCINE. For the treatment of severe anaphylaxis, the initial dose of adrenaline is 0.1 - 0.5 mg (0.1 - 0.5 ml of 1:1000 injection) given s/c or i/m. Single dose should not exceed 1 mg (1ml). For infants and children, the recommended dose of adrenaline is 0.01 mg/kg (0.01 ml/kg of 1:1000 injection). Single paediatric dose should not exceed 0.5 mg (0.5 ml). The mainstay in the treatment of severe anaphylaxis is the prompt use of adrenaline, which can be lifesaving. It should be used at the first suspicion of anaphylaxis.

As with the use of all vaccines, the vaccines should remain under observation for not less than 30 minutes for possibility of occurrence of immediate or early allergic reactions. Hydrocortisone and antihistaminics should also be available in addition to supportive measures such as oxygen inhalation and IV fluids.

Special care should be taken to ensure that the injection does not enter a blood vessel. IT IS EXTREMELY IMPORTANT WHEN THE PARENT, GUARDIAN RETURNS FOR THE NEXT DOSE IN THE SERIES. THE PARENT AND GUARDIAN SHOULD BE QUESTIONED CONCERNING OCCURANCE OF ANY SYMPTOMS AND/OR SIGNS OF AN ADVERSE REACTION AFTER THE PREVIOUS DOSE.

This vaccine should not be given as an intramuscular injection to individuals with thrombocytopaenia or any coagulation disorder, or to those receiving anticoagulant therapy.

This vaccine is not intended to be used for treatment of active infection. As with any vaccine, SII PNEUMOSIL (10-valent) may not protect all individuals receiving the vaccine from pneumococcal disease.

Minor illnesses, such as mild respiratory infection, with or without low grade fever, are not generally contraindications to vaccination. The decision to administer or delay vaccination because of a current or recent febrile illness depends largely on the severity of the symptoms and their etiology. The administration of SII PNEUMOSIL (10-valent) should be postponed in subjects suffering from acute severe febrile illness.

Based on experience with use of other pneumococcal conjugate vaccines, for vaccine serotypes protection against otitis media is expected to be lower than protection against invasive disease. As otitis media is caused by many organisms other than pneumococcal serotypes represented in the vaccine, protection against all otitis media is expected to be low.

SPECIAL POPULATIONS

Safety and immunogenicity data on **SII PNEUMOSIL** (10-valent) are not available for children in specific groups at higher risk for invasive pneumococcal disease (e.g. children with congenital or acquired splenic dysfunction, HIV infection, malignancy, nephrotic syndrome). Children in these groups may have reduced antibody response to active immunization due to impaired immune responsiveness. Limited data have demonstrated that other pneumococcal conjugate vaccines induce an immune response in children with HIV, sickle cell disease, and children born prematurely with a safety profile similar to that observed in non-high-risk groups. The use of **SII PNEUMOSIL** (10-valent) in high-risk groups should be considered on an individual basis.
Apnoea in Premature Infants: Based on experience with use of other pneumococcal conjugate vaccines, the potential risk of apnoea and the need for respiratory monitoring for 48-72 hours should be considered when administering the primary immunization series to very premature infants (born ≤ 28 weeks of gestation) and particularly for those with a previous history of respiratory immaturity. As the benefit of vaccination is high in this group of infants, vaccination with **SII PNEUMOSIL** (10-valent) should not be withheld or delayed.

PEDIATRIC USE

SII PNEUMOSIL (10-valent) is not intended for use in children below the age of 6 weeks. The safety and effectiveness in children below the age of 6 weeks has not been established.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Not relevant.

PREGNANCY AND LACTATION

Human data on the use during pregnancy is not available. Therefore, the use of **SII PNEUMOSIL** (10-valent) should be avoided during pregnancy. It is not known if pneumococcal conjugate vaccine, (10-valent) is excreted in human milk.

INTERACTIONS WITH OTHER MEDICAMENTS

SII PNEUMOSIL (10-valent) can be given with any of the following vaccine antigens, either as monovalent or combination vaccines: diphtheria, tetanus, whole-cell pertussis, *Haemophilus influenzae* type b, inactivated or oral poliomyelitis, rotavirus, yellow fever, hepatitis B, measles and rubella. Clinical studies demonstrated that the immune responses and the safety profiles of the administered vaccines were unaffected. Studies with other pneumococcal conjugate vaccines coadministered with mumps, varicella, meningococcal ACWY and rotavirus vaccines have demonstrated that the immune responses of the other pneumococcal conjugate vaccines and the co-administered vaccines were unaffected.
Based on experience with use of other pneumococcal conjugate vaccines, SII PNEUMOSIL can be given concomitantly with acellular pertussis antigen.
In clinical trials, when other pneumococcal conjugate vaccines were given concomitantly but at a different site/route, with rotavirus vaccine or hepatitis A vaccine, no change in the safety profiles for these infants was observed.
Based on experience with use of other pneumococcal conjugate vaccines, data from a post marketing clinical study evaluating the impact of prophylactic use of antipyretics (ibuprofen and paracetamol) on the immune response to **SII PNEUMOSIL** suggest that administration of paracetamol concomitantly or within the same day of vaccination may reduce the immune response to **SII PNEUMOSIL** after the infant series. Responses to the booster dose administered at 12 months were unaffected. The clinical significance of this observation is unknown.
Different injectable vaccines should always be given at different injection-sites. Till date **SII PNEUMOSIL** (10-valent) clinical studies have been conducted in India and The Gambia in toddlers and infants.
In the Gambia Phase ½ study, there was no evidence that administration of **SII PNEUMOSIL** (10-valent) interfered with the immune response to any component of co-administered pentavalent vaccine.
In the Gambia Phase 3 study, non-inferiority of the immune responses induced by EPI vaccines between treatment groups was demonstrated for all EPI vaccines co-administered during the 3-dose primary vaccination series (6 weeks, 10 weeks and 14 weeks) - namely, whole-cell pentavalent vaccine (DTwP-HepB-Hib) oral polio vaccine, inactivated polio vaccine, and oral rotavirus vaccine. Standard EPI vaccines based on the Gambian EPI schedule (measles-rubella vaccine and yellow fever virus vaccine) were co-administered with the booster dose of study vaccine. Non-inferiority of the immune responses was demonstrated for these co-administered EPI vaccines. While there are no known published data on co-administration of other pneumococcal conjugate vaccine with yellow fever virus vaccine, the high seroresponse rate to yellow fever in the **SII PNEUMOSIL** (10-valent) group indicates that **SII PNEUMOSIL** (10-valent) does not interfere with the immune response to yellow fever virus vaccine.

ADVERSE REACTIONS

Summary of the safety profile

Safety assessment of **SII PNEUMOSIL** (10-valent) was based on clinical trials involving the administration of 6,186 doses to 2,216 healthy infants as primary immunisation. Furthermore, 793 children received a booster dose of **SII PNEUMOSIL** (10-valent) following a primary vaccination course. **SII PNEUMOSIL** (10-valent) was administered concomitantly with recommended childhood vaccines, as appropriate. The vast majority of the reactions observed following vaccination were of mild or moderate severity and were of short duration.
In the largest study in infants, the most common adverse reactions observed after primary vaccination were tenderness at the injection site, fever and irritability, which were reported for approximately 49%, 52% and 32% of all infants, respectively. No increase in the incidence or severity was observed following subsequent doses of the primary vaccination course. Following booster vaccination, the most common adverse reaction was tenderness at the injection site, which was reported for approximately 8% of all infants.
The Indian Phase 3 licensure study in infants similarly showed tenderness at the injection site, fever and irritability as the most common adverse reactions observed after primary vaccination, with no change in the incidence or severity observed following subsequent doses of the primary vaccination course. Majority of the solicited AEs were of mild to moderate intensity and resolved completely.
In the Gambian Phase 3 descriptive study in infants, the most common adverse reaction observed were tenderness at the injection site, irritability and fever, which were reported for approximately 31.8%, 47.7% and 44.5% of all infants respectively. Majority of the solicited AEs were of mild to moderate intensity and resolved completely.
The Indian Phase 3 descriptive study in infants similarly showed tenderness at the injection site, fever and irritability as the most common adverse reactions observed after vaccination. Majority of the solicited AEs were of mild to moderate intensity and resolved completely.
Safety was also assessed in 57 previously unvaccinated children during the second year of life; all children received 2 doses of vaccine. Pneumococcal Polysaccharide Conjugate Vaccine (Adsorbed) (10-valent) has also been used for booster vaccination in 56 children who received another pneumococcal conjugate vaccine for the primary course.
The injection site and systemic reactions following catch-up vaccination or booster vaccination during the second year of life were comparable to those reported after primary vaccination.
In all studies, the incidence and severity of local and general adverse reactions reported within 7 days of vaccination were comparable to those after vaccination with the licensed comparator PCVs

Tabulated list of adverse reactions

Adverse reactions (i.e. events considered as related to vaccination) have been categorised by frequency for all age groups.
Frequencies are reported as:
Very common (≥1/10 vaccinees)
Common (≥1/100 vaccinees but < 1/10 vaccinees)
Uncommon (≥1/1000 vaccinees but < 1/100 vaccinees)
Rare (≥1/10,000 vaccinees but < 1/1,000 vaccinees)

System Organ Class	Frequency	Adverse reactions
Gastrointestinal disorders	Uncommon	Diarrhoea
General disorders and administration site conditions	Very common	Pain, Fever ⩾ 37.5°C (axillary)
	Common	Erythema, Swelling/induration
	Uncommon	Fever > 39°C (axillary)
Metabolism and nutrition disorders	Common	Decreased appetite
Nervous system disorders	Common	Drowsiness
Psychiatric disorders	Very common	Irritability
Skin and subcutaneous tissue disorders	Common	Rash

Based on experience with use of other pneumococcal conjugate vaccines, analysis of post marketing reporting rates suggests a potential increased risk of convulsions, with or without fever, and HHE when comparing groups which reported use of **SII PNEUMOSIL** with Infanrix hexa to those which reported use of **SII PNEUMOSIL** alone.

PHARMACODYNAMICS

Pharmacodynamic properties:
Pharmacotherapeutic group: Vaccines
Pneumococcal Polysaccharide Conjugate Vaccine (Adsorbed) (10-Valent) (**SII PNEUMOSIL**),
ATC code J07AL02.
Immunological Data:
Clinical trials performed to assess immunogenicity and reactogenicity of the vaccine and proved that the vaccine is immunogenic.

PHARMACOKINETICS

Pharmacokinetic studies are not required for vaccines.

PRECLINICAL SAFETY DATA

Single and multiple administration of the **SII PNEUMOSIL** (10-valent) to rats and rabbits were well tolerated and revealed no evidence of any significant local or systemic toxic effects. Observed changes were not considered adverse but rather a consequence of the pharmacological activity of **SII PNEUMOSIL** (10-valent).

COMPATIBILITIES/INCOMPATIBILITIES

The vaccine is not to be mixed with other vaccines/products in the same syringe.

SYMPTOMS AND TREATMENT OF OVERDOSE

No case of overdose has been reported.

STORAGE

Store in a refrigerator (2°C – 8°C).
DO NOT FREEZE. Discard if the vaccine has been frozen.

A fine white deposit with clear colourless supernatant may be observed upon storage of the vial. This does not constitute a sign of deterioration.

For 5-dose (Multi-dose)
After first opening of the 5-dose vial, the vaccine may be stored for a maximum of 28 days in a refrigerator (2°C – 8°C). If not used within 28 days it should be discarded.

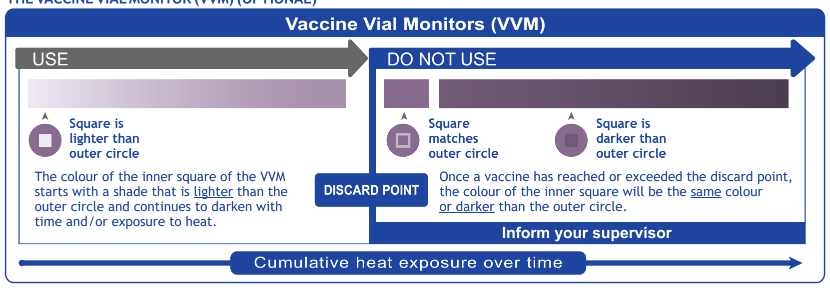
SHELF LIFE

36 months from the date of manufacture.

PRESENTATION

1 dose - 0.5 ml vial. Pack of 1 vial, 10 vials.
5 dose - 2.5 ml vial. Pack of 50 vials.

THE VACCINE VIAL MONITOR (VVM) (OPTIONAL)



Vaccine Vial Monitors (VVMs) are on the cap of the vial / part of the label on **SII PNEUMOSIL** (10-valent) supplied through Serum Institute of India Pvt. Ltd. This is a time-temperature sensitive dot that provides an indication of the cumulative heat to which the vial has been exposed. It warns the end user when exposure to heat is likely to have degraded the vaccine beyond an acceptable level.
The interpretation of the VVM is simple. Focus on the central square. Its colour will change progressively. As long as the colour of this square is lighter than the colour of the outer circle, then the vaccine can be used. As soon as the colour of the central square is the same colour as the outer circle or of a darker colour than the outer circle, then the vial should be discarded.

PRODUCT REGISTRATION HOLDER

Pharmaniaga LifeScience Sdn Bhd (198201002939)
Lot 7, Jalan PPU 3,
Taman Perindustrian Puchong Utama,
47100 Puchong, Selangor Darul Ehsan, Malaysia.

Manufactured and released by:
SERUM INSTITUTE OF INDIA PVT. LTD.
212/2, Hadapsar, Pune 411 028, INDIA

Revision date: 03/2023

™ Trademark under Registration

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Supercedes Item Code: 20017103/2	Dimensions: 123 x 373 mm	

PACKAGING DEVELOPMENT	QUALITY CONTROL	REGULATORY AFFAIRS	MEDICAL DEPARTMENT	QUALITY ASSURANCE

File Name: E:\Packaging artworks as on 090212\Artworks SII Pvt Ltd 231015\Insert\Malaysia\PNEUMOSIL Insert, Malaysia.cdr Rev. on: 07.03.2023